

Curriculum Vitae

Johannes Köster

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Personal information

Homepage: <https://koesterlab.github.io>
Birth: May 5, 1985 in Gelsenkirchen, Germany
Nationality: German
Family status: Married, two children

Positions

Current positions

since 2023: **Full professor (W3, tenure track)** for Bioinformatics and Computational Oncology (Lehrstuhl Bioinformatische Algorithmen in der Onkologie) at the University of Duisburg-Essen.
since 2020: **DKTK Investigator** at the German Consortium of Translational Cancer Research (DKTK) of WTZ and DKFZ.
since 2023: **Scientific coordinator** for the core “computational biomedicine” in the BIOME graduate school of the University of Duisburg-Essen.
since 2016: **Affiliated consulting scientist** in Myles Brown's group (Division of Molecular and Cellular Oncology, Department of Medical Oncology, Dana-Farber Cancer Institute, Harvard Medical School), USA.

Previous positions

2017 – 2023: **Group leader** (permanent position), Algorithms for reproducible bioinformatics (<https://koesterlab.github.io>), Genome Informatics, Institute of Human Genetics, Faculty of Medicine, University of Duisburg-Essen. Currently eight group members (graduate students).
2016 – 2017: **Researcher** (NWO Veni funded, independent) at Life Sciences, Centrum Wiskunde & Informatica, Amsterdam, Netherlands.
2016: **Postdoctoral research fellow** under the supervision of Alexander Schönhuth, Life Sciences, Centrum Wiskunde & Informatica, Amsterdam, Netherlands.
2016: **Consultant** (data analysis and workflow management) for Juno therapeutics, 307 Westlake Avenue, Seattle, USA.
2015 – 2016: **Postdoctoral research fellow** in Xiaole Shirley Liu's group (Department of Biostatistics and Computational Biology, Dana-Farber Cancer Institute, Harvard School of Public Health) and Myles Brown's group (Division of Molecular and Cellular Oncology, Department of Medical Oncology, Dana-Farber Cancer Institute, Harvard Medical School), USA.
2011 – 2015: **Research fellow** at both the Chair of Genome Informatics of Sven Rahmann, University of Duisburg-Essen and in the “European Network for Cancer in Children and Adolescents” project under Alexander Schramm, University Hospital Essen, Germany.

Education

2011-2015: **Ph.D. in Computer Science (grade “ausgezeichnet”/summa cum laude)** at TU Dortmund, Germany. Thesis title: “Parallelization, Scalability, and Reproducibility in Next-Generation Sequencing Analysis”. Supervisor: Sven Rahmann, Chair of Genome Informatics, University of Duisburg-Essen. Second assessor: Axel Mosig, Bioinformatics Group, Ruhr University Bochum.
2005 – 2010: **Diploma (equiv. to M.Sc.) in Computer Science (grade 1.0/A with honors)** at TU Dortmund, Germany. Thesis title: “Propagating Interaction Logic toward Predictive Protein Hypernetworks”. Supervisors: Sven Rahmann, Bioinformatics for High-Throughput-Technologies, Computer Science XI, TU Dortmund and Eli Zamir, Systems Biology of Cell-Matrix Adhesion, Department of Systemic Cell Biology, Max Planck Institute of Molecular Physiology Dortmund.

Awards

- 2019: Friedmund-Neumann-Prize** for outstanding fundamental research on reproducibility in biomedicine. Schering Foundation. Amount: 10,000 €
- 2011: Hans-Uhde-Prize** by the ThyssenKrupp Uhde GmbH for extraordinary diploma thesis with practical application in Computer Science. Amount: 500 €
- 2021: Young scientist best paper award** of the Faculty of Medicine, University of Duisburg-Essen for the paper “Varlociraptor: enhancing sensitivity and controlling false discovery rate in somatic indel discovery” .
- 2019: Young scientist best paper award** of the Faculty of Medicine, University of Duisburg-Essen for the paper “Bioconda: sustainable and comprehensive software distribution for the life sciences” .
- 2014: NVIDIA Hardware Grant** received for publication “Massively parallel read mapping on GPUs with the q-group index and PEANUT”. Amount: 2000 € (Tesla GPU)

Organized Events

- Sep 2026 PC member:** German Conference on Bioinformatics (GCB) 2026
- Sep 2025 PC member:** German Conference on Bioinformatics (GCB) 2025
- Dec 2022 Session chair:** Biological data science, Cold Spring Harbor labs 2022
- Nov 2021 PC member:** Computational Intelligence Methods for Bioinformatics and Biostatistics (CIBB) 2021
- Feb 2019 Co-Organizer:** Data structures in Bioinformatics (DSB) 2019, Dortmund, Germany
- Jun 2018 Co-Organizer:** Lorentz workshop, Leiden, The Netherlands: Making sense of millions of single cells: approaching the era of Single Cell Data Science, experimental advances and computational challenges
- Sep 2017 PC member:** CIBB 2017, Cagliari, Italy
- Mar 2017 Organizer:** Tutorial “Reproducible data analysis with Snakemake” at CWI Amsterdam
- Feb 2017 Co-Organizer and session chair:** workshop “data structures in bioinformatics” (DSB) 2017 (<https://dsb2017.github.io>)

Invited talks

- May 2026 University of Cologne:** Transparent, adaptable, and reproducible data analyses for oncology - Snakemake, Datavzrd, Vembrane and beyond
- Feb 2026 CERN:** Snakemake in 2026
- Sep 2025 EMBL:** Snakemake in 2025
- Apr 2025 German Human Genome Phenome Archive (GHGA):** Benchmarking genomic variant calling
- Jan 2025 BASF:** Reproducible data analysis with Snakemake
- Oct 2024 University of Torino:** Interactive data analysis with Snakemake and Jupyter
- Sep 2024 German Conference on Bioinformatics (GCB):** Generic solutions for specific problems (keynote)
- Dec 2022 Biological data science, Cold Spring Harbor labs:** Rapidly configurable, portable, interactive visualization of arbitrary tabular results
- Feb 2022 Cambridge Genetics Seminar:** Reproducibility of NGS data analysis
- Dec 2021 Helmholtz Center Dresden:** Sustainable data analysis
- Jul 2021 Intelligent Systems for Molecular Biology Conference (ISMB), BIOINFO-CORE track:** Keynote on Snakemake
- Aug 2020 University Medicine Mainz:** Reproducibility and transparency in bioinformatics
- May 2019 NIH:** Reproducible data analysis with Snakemake
- May 2019 Harvard University:** Towards a unified theory of variant calling
- Feb 2019 Sanger Institute:** Reproducible data analysis with Snakemake
- Dec 2018 Martin Luther University Halle-Wittenberg:** Reproducible data analysis with Snakemake
- Sep 2018 Bernstein Conference on computational neuroscience, Berlin:** Reproducible data analysis with Snakemake
- Nov 2017 Robert-Koch-Institute, Berlin:** Benchmarking RNA-seq data analysis
- Feb 2017 Gutenberg University Mainz:** Snakemake and Bioconda
- Feb 2017 Forschungszentrum Jülich:** Snakemake and Bioconda
- Nov 2016 Helmholtz Open Science Workshop, Dresden, Germany:** Keynote about sustainable software development in science.
- Sep 2016 Netherlands Cancer Institute (NKI), Amsterdam:** Snakemake and Bioconda.
- Jul 2016 Rust bay area meeting, Mozilla Foundation Los Angeles:** Bioinformatics with Rust.
- Jun 2016 Aquatic Ecosystem Research, University of Duisburg-Essen:** Snakemake and Bioconda.
- Jun 2016 Leiden University Medical Center, Netherlands:** Algebraic variant calling, Snakemake and Bioconda.
- Jul 2015 Broad Institute, Boston, USA:** Workflow Management with Snakemake

Apr 2014 **Workshop on Fast Data Processing on GPUs, CUDA Center of Excellence, Dresden:** Massively parallel read mapping on graphics cards.

Apr 2014 **DTL Focus Meeting, Utrecht, Netherlands:** Reproducible data analysis with Snakemake

Teaching

- SS 2024** **Praxisprojekt**¹ Reinventing email, Faculty of Computer Science, University of Duisburg-Essen (6-12 SWS)
- SS 2025** **Seminar** Data Analysis and Visualization, Faculty of Medicine, University of Duisburg-Essen (2 SWS)
- SS 2024** **Seminar** Data Analysis and Visualization, Faculty of Medicine, University of Duisburg-Essen (2 SWS)
- SS 2024** **Praxisprojekt** Implementation of algorithms and data structures with Rust, Faculty of Computer Science, University of Duisburg-Essen (6 SWS)
- WS 2023** **Praxisprojekt** Implementation of algorithms and data structures with Rust, Faculty of Computer Science, University of Duisburg-Essen (6 SWS)
- SS 2019** **Fachprojekt** "Algorithm Engineering: Entwicklung einer Rust-Bibliothek am Beispiel von Wavelet Trees", Faculty of Computer Science, TU Dortmund (4 SWS)
- WS 2018** **Fachprojekt** "Bioinformatik: Reproduzierbare Datenanalyse mit Snakemake am Beispiel der Bioinformatik", Faculty of Computer Science, TU Dortmund (4 SWS)
- SS 2018** **Fachprojekt** "Algorithm Engineering: Entwicklung einer Rust-Bibliothek am Beispiel von Succinct Trees", Faculty of Computer Science, TU Dortmund (4 SWS)
- since 2016** **21 invited tutorials** on reproducible data analysis with Snakemake (duration: 1 or 2 days; Sanger Institute, Harvard University, NIH, ETH Zürich, Berlin Institute of Health, Humboldt University Berlin, Norwegian University of Science and Technology, CWI Amsterdam, Universiteit Utrecht, INRA Toulouse, University of Duisburg-Essen, Wageningen University, AstraZeneca, RTG WisPerMed, Physalia Courses, Vlaams Instituut voor Biotechnologie).
- SS 2013** **One guest lecture** about variant calling with next-generation sequencing in the course "Statistic in Genetics", Faculty of Statistics, TU Dortmund, Germany.
- WS 2012** **Two guest lectures** about protein networks, complex prediction and interaction dependencies in the course "Computational Omics", Faculty of Computer Science, TU Dortmund, Germany.
- WS 2011** **Teaching assistant** for "Datastructures, Algorithms and Programming 1", at the Chair of Computer Science X, TU Dortmund, Germany. (2 SWS)
- 2007 – 2009** **Student teaching assistant** for "Datastructures, Algorithms and Programming 1", at the Chair of Computer Science X, TU Dortmund, Germany. (2 SWS)

Reviewing

- since 2022** Grant application reviewer for Biotechnology and Biological Sciences Research Council (BBSRC), UK
- since 2021** Grant application reviewer for National Science Centre (NCN), Poland
- since 2024** Grant application reviewer for German Ministry of Research (BMBF)
- since 2012** Reviewer for various journals (Nature, Nature Biotechnology, Nature Communications, Nature Methods, Bioinformatics, BMC Bioinformatics, Genome Biology, Cell Systems, PLOS Computational Biology, GigaScience, PeerJ)

Original publications

1. Wiegand F, Mölder F, **Köster J**. Alignoth: portable and interactive visualization of read alignments. *Bioinformatics*. 2026 Jan 1;42(1):btaf663.
2. Uzuner H, Wiegand F, Schrinner S, Lähnemann D, Schadendorf D, **Köster J**. Virus variant quantification with Orthanq. *BMC Bioinformatics*. 2026 Feb 4;27(1):49.
3. Gregor K, Meyer BF, Gaida T, Justo Vasquez V, Bett-Williams K, Forrest M, Darel-Filho JP, Rabin S, Longo M, Melton JR, Nord J, Anthoni P, Bastrikov V, Colligan T, Delire C, Dietze MC, Hurtt G, Ito A, Keetz LT, Knauer J, **Köster J**, Lin TS, Ma L, Minvielle M, Olin S, Ostberg S, Shi H, Schnur R, Sun Q, Thornton PE, Rammig A. Best practices in software development for robust and reproducible geoscientific models based on insights from the Global Carbon Budget's dynamic vegetation models. *Geoscientific Model Development*. 2026 Mar 25;19(6):2407–36.
4. Wilkinson SR, Aloqalaa M, Belhajjame K, Crusoe MR, de Paula Kinoshita B, Gadelha L, Garijo D, Gustafsson OJR, Juty N, Kanwal S, Khan FZ, **Köster J**, Peters-von Gehlen K, Pouchard L, Rannow RK, Soiland-Reyes S, Soranzo N, Sufi S, Sun Z, Vilne B, Wouters MA, Yuen D, Goble C. Applying the FAIR Principles to computational workflows. *Sci Data*. 2025 Feb 24;12(1):328.
5. Wiegand F, Lähnemann D, Mölder F, Uzuner H, Prinz A, Schramm A, **Köster J**. Datavzrd: Rapid programming- and maintenance-free interactive visualization and communication of tabular data. *PLoS One*. 2025;20(7):e0323079.
6. Rausch T, Schäfer H, Bein J, Mölder F, Beeck LK, Kölsch B, Borchert S, Benes V, Halbsguth T, Brunnberg U, Oellerich T, Tousseyn T, Ponzoni M, **Köster J**, Hansmann ML, Küppers R, Hartmann S. Genetic lesions in nodular lymphocyte-predominant Hodgkin lymphoma and T cell/histiocyte-rich large B-cell lymphoma identified by whole genome sequencing. *Leukemia*. 2025 Sept;39(9):2215–25.

¹A "Fachprojekt/Praxisprojekt" is a hybrid 1-semester course in which students solve a computer science problem in small groups. It thereby consists of lecture, exercise, teamwork, and seminar components.

7. Machlah YM, Brandenburg T, Hönes GS, Theurer S, Prinz AD, Hoppe C, Cansiz F, Schulte JH, Kero J, Undeutsch H, Siveke J, **Köster J**, Fuehrer D, Moeller LC. ALK Inhibition Prolongs Survival in a Mouse Model of ALK-Positive Anaplastic Thyroid Cancer. *Thyroid*. 2025 Dec 30;10507256251409070.
8. Uzuner H, Paschen A, Schadendorf D, **Köster J**. Orthanc: transparent and uncertainty-aware haplotype quantification with application in HLA-typing. *BMC Bioinformatics*. 2024 July 16;25(1):240.
9. Schuler M, Cuppens K, Plönes T, Wiesweg M, Du Pont B, Hegedus B, **Köster J**, Mairinger F, Darwiche K, Paschen A, Maes B, Vanbockrijck M, Lähmann D, Zhao F, Hautzel H, Theegarten D, Hartemink K, Reis H, Baas P, Schramm A, Aigner C. Neoadjuvant nivolumab with or without relatlimab in resectable non-small-cell lung cancer: a randomized phase 2 trial. *Nat Med*. 2024 June;30(6):1602–11.
10. Hanssen F, Gabernet G, Bäuerle F, Stöcker B, Wiegand F, Smith NH, Mertes C, Neogi AG, Brandhoff L, Ossowski A, Altmueller J, Becker K, Petzold A, Sturm M, Stöcker T, Sivalingam S, Brand F, Schmidt A, Buness A, Probst AJ, Motameny S, **Köster J**. NCBench: providing an open, reproducible, transparent, adaptable, and continuous benchmark approach for DNA-sequencing-based variant calling. *F1000Research* [Internet]. 2024 Sept 12 [cited 2024 Nov 15];12:1125. Available from: <https://f1000research.com/articles/12-1125>
11. Hartmann T, Schröder C, Kuthe E, Lähmann D, **Köster J**. Insane in the vembrane: filtering and transforming VCF/BCF files. *Bioinformatics*. 2023 Jan 1;39(1):btac810.
12. Bergkamp ND, van Senten JR, Brink HJ, Bebelman MP, van den Bor J, Çobanoğlu TS, Dinkla K, **Köster J**, Klau G, Siderius M, Smit MJ. A virally encoded GPCR drives glioblastoma through feed-forward activation of the SK1-S1P1 signaling axis. *Science Signaling*. 2023 Aug 15;16(798):eade6737.
13. Tüns, A.I., Hartmann, T., Magin, S., González, R.C., Henssen, A.G., Rahmann, S., Schramm, A., **Köster, J. (shared last author)** Detection and Validation of Circular DNA Fragments Using Nanopore Sequencing. *Frontiers in Genetics* 13 (2022).
14. Mölder, F., Jablonski, K.P., Letcher, B., Hall, M.B., Tomkins-Tinch, C.H., Sochat, V., Forster, J., Lee, S., Twardziok, S.O., Kanitz, A., Wilm, A., Holtgrewe, M., Rahmann, S., Nahnsen, S., **Köster, J (corresponding author)**. Sustainable data analysis with Snakemake. *F1000Res* 10, 33 (2021).
15. Cormier, M.J., Belyeu, J.R., Pedersen, B.S., Brown, J., **Köster, J.**, Quinlan, A.R. Go Get Data (GGD) is a framework that facilitates reproducible access to genomic data. *Nat Commun* 12, 2151 (2021).
16. Wiesweg, M., Mairinger, F., Reis, H., Goetz, M., Kollmeier, J., Misch, D., Stephan-Falkenau, S., Mairinger, T., Walter, R.F.H., Hager, T., Metzenmacher, M., Eberhardt, W.E.E., Zaun, G., **Köster, J.**, Stuschke, M., Aigner, C., Darwiche, K., Schmid, K.W., Rahmann, S., Schuler, M. Machine learning reveals a PD-L1 independent prediction of response to immunotherapy of non-small cell lung cancer by gene expression context. *European Journal of Cancer* 140, 76–85 (2020).
17. Such, L., Zhao, F., Liu, Derek, Thier, B., Le-Trilling, V.T.K., Sucker, A., Coch, C., Pieper, N., Howe, S., Bhat, H., Kalkavan, H., Ritter, C., Brinkhaus, R., Ugurel, S., **Köster, J.**, Seifert, U., Dittmer, U., Schuler, M., Lang, K.S., Kufer, T.A., Hartmann, G., Becker, J.C., Horn, S., Ferrone, S., Liu, David, Allen, E.M.V., Schadendorf, D., Griewank, K., Trilling, M., Paschen, A., Targeting the innate immunoreceptor RIG-I overcomes melanoma-intrinsic resistance to T cell immunotherapy. *J Clin Invest* 130, 4266–4281 (2020).
18. **Köster, J. (corresponding author)**, Dijkstra, L. J., Marschall, T., Schönhuth, A. Varlociraptor: enhancing sensitivity and controlling false discovery rate in somatic indel discovery. *Genome Biology* 21, 98 (2020).
19. Baaijens, J.A., Van der Roest, B., **Köster, J.**, Stougie, L., Schönhuth, A. Full-length de novo viral quasiespecies assembly through variation graph construction. *Bioinformatics* 35, 5086–5094 (2019).
20. **Köster (corresponding author)**, J., Brown, M., Liu, X.S. A Bayesian model for single cell transcript expression analysis on MERFISH data. *Bioinformatics* 35, 995–1001 (2019).
21. Grüning, B., [...], **Köster, J.** et al. Practical Computational Reproducibility in the Life Sciences. *Cell Systems* 6, 631–635 (2018).
22. Grüning, B., Dale, R., Sjödin, A., Chapman, B.A., Rowe, J., Tomkins-Tinch, C.H., Valieris, R., the Bioconda Team, **Köster, J. (corresponding author)**. Bioconda: sustainable and comprehensive software distribution for the life sciences. *Nat Methods* 15, 475–476 (2018).
23. Cornwell, M., Vangala, M., Taing, L., Herbert, Z., **Köster, J.**, Li, B., Sun, H., Li, T., Zhang, J., Qiu, X., Pun, M., Jeselsohn, R., Brown, M., Liu, X. S., and Long, H. VIPER: Visualization Pipeline for RNA-seq, a Snakemake workflow for efficient and complete RNA-seq analysis. *BMC Bioinformatics* 19:135 (2018).
24. Schulte, M., **Köster, J.**, Rahmann, S., Schramm, A. Cancer evolution, mutations, and clonal selection in relapse neuroblastoma. *Cell Tissue Res* 372:2 (2018).
25. Stöcker, B. K., **Köster, J.**, Rahmann, S. SimLoRD: Simulation of Long Read Data. *Bioinformatics*, 32(17), 2704–2706 (2016).
26. Ma, J., **Köster, J.**, Qin, Q., Hu, S., Li, W., Chen, C., Cao, Q., Wang, J., Mei, S., Liu, Q., Xu, H., Liu, X.S. CRISPR-DO for genome-wide CRISPR design and optimization. *Bioinformatics* 32, 3336–3338 (2016).
27. Townsend, E. C., Murakami, M. A., Christodoulou, A., Christie, A. L., **Köster, J. (first collaborative author)**, DeSouza, T. A., ... Weinstock, D. M. The Public Repository of Xenografts Enables Discovery and Randomized Phase II-like Trials in Mice. *Cancer Cell*, 29(4), 574–586 (2016).
28. Li, W., **Köster, J. (co-first author)**, Xu, H., Chen, C.-H., Xiao, T., Liu, J. S., ... Liu, X. S. Quality control, modeling, and visualization of CRISPR screens with MAGeCK-VISPR. *Genome Biology*, 16(1), 281 (2015).
29. **Köster, J.** Rust-Bio: a fast and safe bioinformatics library. *Bioinformatics* 32, 444–446 (2016).
30. Schramm, A., **Köster, J.**, Assenov, Y., Althoff, K., Peifer, M., Mahlow, E., Odersky, A., Beisser, D., Ernst, C., Henssen, A. G., Stephan, H., Schröder, C., Heukamp, L., Engesser, A., Kahlert, Y., Theissen, J., Hero, B., Roels, F., Altmüller, J., Nürnberg, P., Astrahantseff, K., Gloeckner, C., De Preter, K., Plass, C., Lee, S., Lode, H. N., Henrich, K., Gartlgruber, M., Speleman, F., Schmezer, P., Westermann, F., Rahmann, S., Fischer, M., Eggert, A., Schulte, J. H. Mutational dynamics between primary and relapse neuroblastomas. *Nature Genetics*, 47(8), 872–877 (2015).
31. Schwermer, M., Lee, S., **Köster, J.**, Maerken, T. van, Stephan, H., Eggert, A., Morik, K., Schulte, J. H., Schramm, A., van Maerken, T. Sensitivity to cdk1-inhibition is modulated by p53 status in preclinical models of embryonal tumors. *Oncotarget*, 6(17), 15425–15435 (2015).
32. **Köster, J. (corresponding author)**, Rahmann, S. Massively parallel read mapping on GPUs with the q-group index and PEANUT. *PeerJ* 2, e606 (2014).

33. Schramm, A., **Köster, J.**, Marschall, T., Martin, M., Schwermer, M., Fielitz, K., Büchel, G., Barann, M., Esser, D., Rosenstiel, P., Rahmann, S., Eggert, A., Schulte, J. H. Next-generation RNA sequencing reveals differential expression of MYCN target genes and suggests the mTOR pathway as a promising therapy target in MYCN-amplified neuroblastoma. *International Journal of Cancer* 132, E106–E115 (2013).
34. Rahmann, S., Martin, M., Schulte, J. H., **Köster, J.**, Marschall, T., Schramm, A. Identifying transcriptional miRNA biomarkers by integrating high-throughput sequencing and real-time PCR data. *Methods* 59, 154–163 (2013).
35. Althoff, K., Beckers, A., Odersky, A., Mestdagh, P., **Köster, J.**, Bray, I.M., Bryan, K., Vandesompele, J., Speleman, F., Stallings, R. L., Schramm, A., Eggert, A., Sprüssel, A., Schulte, J. H. MiR-137 functions as a tumor suppressor in neuroblastoma by downregulating KDM1A. *International Journal of Cancer* 133, 1064–1073 (2013).
36. **Köster, J. (corresponding author)**, Rahmann, S. Snakemake – a scalable bioinformatics workflow engine. *Bioinformatics* 28, 2520–2522 (2012).
37. **Köster, J.**, Zamir, E., Rahmann, S. Efficiently mining protein interaction dependencies from large text corpora. *Integr. Biol.* 4, 805–812 (2012).
38. **Köster, J. (corresponding author)**, Rahmann, S. Building and Documenting Workflows with Python-Based Snakemake, in: Böcker, S., Hufsky, F., Scheubert, K., Schleicher, J., Schuster, S. (Eds.), *German Conference on Bioinformatics 2012, OpenAccess Series in Informatics (OASISs)*. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, Dagstuhl, Germany, pp. 49–56 (2012).
39. Schramm, A., Schowe, B., Fielitz, K., Heilmann, M., Martin, M., Marschall, T., **Köster, J.**, Vandesompele, J., Vermeulen, J., de Preter, K., Koster, J., Versteeg, R., Noguera, R., Speleman, F., Rahmann, S., Eggert, A., Morik, K., Schulte, J. H. Exon-level expression analyses identify MYCN and NTRK1 as major determinants of alternative exon usage and robustly predict primary neuroblastoma outcome. *Br J Cancer* 107, 1409–1417 (2012).

Review publications

1. David Lähnemann, **Johannes Köster (shared first and last author)**, Ewa Szczurek, Davis J. McCarthy, Stephanie C. Hicks, Mark D. Robinson, Catalina A. Vallejos, Kieran R. Campbell, et al. Eleven grand challenges in single-cell data science. *Genome Biology* 21, 31 (2020).